Midterm Exam Machine learning.

Last Name: First Name:

1. Explain differences between supervised and unsupervised learning (20 p) (2 or 3 lines is enough)
2. Describe the relation between training error and overfitting/underfitting of the machine learning model. (20 p)
3. Considering the following observations:

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Calculate the resulting class (Yes or No} that a Bayesian classifier would return for the following observation: x'=(Outlook=Sunny, Temperature=Cool, Humidity=High, Wind=Weak)

Build the learning phase and the test phase. (Help from supplemental material) (30 p).

4. Assuming that the RSS (a,b) get to be reduced to RSS(a,b)=a^2+b^2−4a−6b+13 . Calculate the best fitting linear model with the (gradient=0) method. (30 p).

Troy University Resources